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Assignment 8: Weighted Interval Scheduling Problem

1. Problem description/comments:

Problem: Given a set of intervals, find the maximum total value of a set of non-overlapping intervals

The issue i sto break a problem (findina maximum) into sub-problems, and build the result from the itervals that have the constranint that they do not overlap.

The overlap is determined by comparing the end times of the interavals; two intervals do not overlap if the end time of the first is less than or equal to the start time of the second. For example, the intervals (2, 6) and (5, 9) overlap because the first goes from time 2 to 6, and the second starts at 5, before the first has finished.

2. Tests

Test data is included in the code, with different blocks that can be uncommented, along with the expected results that they should create.

Included:
- Normal intervals, mixed overlap
- A "greedy fails" situation
- A case where all intervals are compatible
- A case where NO two intervals are compatible
- An "edge case", with only a single interval

3. Implementation

See included code files. (weighted.js)

4. Reflection

What is the state in this problem?
The number of intervals from the sorted list

What does p(j) mean?
The closest previous non-overlapping interval.

What is the recurrence?
Maximizing the total values of the last J intervals

How is this problem similar to House Robber, and how is it more complex?
They share the same basic logic, at each step, you decide whether to take or skip the item, then combine the results. But with weighted intervals, you can only combine the intervals when they are compatible. Thus it's more complex.